

Partial result for Question 3 of Cuth–Dolezal–Doucha–Kurka

Almost-Hilbertian subspaces force the isometric homogeneous space to be Hilbert

status: partial result, not a full solution

Source Question

The source is Cuth–Dolezal–Doucha–Kurka, *Polish spaces of Banach spaces. Complexity of isometry and isomorphism classes*, arXiv:2204.06834v1. In Section 6, Question 3 asks:

Let X be a separable infinite-dimensional Banach space which is isometric to all of its closed infinite-dimensional subspaces. Is then X isometric to ℓ_2 ?

Solving the homogeneous Banach space problem, Komorowski and Tomczak-Jaegermann ([35]), and Gowers ([29]) proved that if a separable infinite-dimensional Banach space is isomorphic to all of its closed infinite-dimensional subspaces, then it is isomorphic to ℓ_2 . It seems that the isometric variant of this result is open; that is, whether ℓ_2 is the only (up to isometry) separable infinite-dimensional Banach space that is isometric to all of its infinite-dimensional closed subspaces. We note that any Banach space satisfying this criterion must be, by the Gowers' result, isomorphic to ℓ_2 . Our initial interest in this problem was that we observed that a positive answer implies that whenever $\langle X \rangle_\equiv$ is closed in $\mathcal{P}_\infty$ then $X \equiv \ell_2$. Eventually we found another argument (see Subsection 2.1), but the question is clearly of independent interest.

Question 3. Let X be a separable infinite-dimensional Banach space which is isometric to all of its closed infinite-dimensional subspaces. Is then X isometric to ℓ_2 ?

We note here that Question 3 was already attacked by N. de Rancourt [15], who was able to prove that if X is as above (that is, isometric to all of its closed infinite-dimensional subspaces) then, for every $\varepsilon > 0$, X admits a $(1 + \varepsilon)$ -unconditional basis.

Definitions

Call a Banach space X *isometrically homogeneous* if every closed infinite-dimensional subspace of X is linearly isometric to X .

By the homogeneous Banach space theorem of Komorowski–Tomczak-Jaegermann and Gowers, as cited in the source paper, an isometrically homogeneous separable Banach space is linearly isomorphic to ℓ_2 . Thus its norm may be viewed as an equivalent norm on a separable Hilbert space. An equivalent Hilbert norm on the underlying vector space will be denoted by $|\cdot|$.

Idea of the Proof

The source question is hard only after one allows genuinely distortive equivalent Hilbert norms. If the given norm has an infinite-dimensional restriction which is arbitrarily close to a scalar multiple of some Hilbert norm, then the isometric homogeneity transfers that almost-Hilbert structure back to the whole space. Consequently every finite-dimensional subspace of X has Banach–Mazur distance 1 from a Euclidean space. Finite-dimensional compactness then makes every finite-dimensional subspace exactly Euclidean, and the parallelogram law gives a Hilbert norm.

The Partial Result

Theorem 1. *Let $X = (X, \|\cdot\|)$ be a separable infinite-dimensional Banach space which is linearly isometric to all of its closed infinite-dimensional subspaces. Suppose that for some equivalent Hilbert norm $|\cdot|$ on the underlying vector space, the following stabilization property holds: for every $\varepsilon > 0$, there are a closed infinite-dimensional subspace $Y \subset X$ and a scalar $a > 0$ such that*

$$a |y| \leq \|y\| \leq (1 + \varepsilon) a |y| \quad (y \in Y).$$

Then X is linearly isometric to ℓ_2 .

Proof. Fix a finite-dimensional subspace $F \subset X$ and $\varepsilon > 0$. By the assumed stabilization property, choose a closed infinite-dimensional subspace $Y \subset X$ and a scalar $a > 0$ such that

$$a |y| \leq \|y\| \leq (1 + \varepsilon) a |y| \quad (y \in Y).$$

Since X is isometrically homogeneous, there is a surjective linear isometry $U : X \rightarrow Y$ for the norm $\|\cdot\|$. Hence, for every $x \in F$,

$$a |Ux| \leq \|Ux\| = \|x\| \leq (1 + \varepsilon) a |Ux|.$$

The range $U(F)$, equipped with $|\cdot|$, is a finite-dimensional Hilbert space. Therefore F has Banach–Mazur distance at most $1 + \varepsilon$ from a Euclidean space of dimension $\dim F$.

As $\varepsilon > 0$ was arbitrary, $d(F, \ell_2^{\dim F}) = 1$. In finite dimension, Banach–Mazur distance 1 from Euclidean space implies linear isometry to Euclidean space: one may take a convergent subsequence of almost-isometries in the compact unit sphere of operators between fixed finite-dimensional spaces, or equivalently use compactness of the finite-dimensional Banach–Mazur compactum. Thus every finite-dimensional subspace $F \subset X$ is Euclidean.

In particular, every two-dimensional subspace of X is Euclidean, so the parallelogram identity holds for every pair $x, z \in X$. By the Jordan–von Neumann characterization, the norm on X is induced by an inner product. Since X is separable and infinite-dimensional, it is linearly isometric to ℓ_2 . $\square$

Corollary 1. *If Question 3 has a non-Hilbert counterexample X , then for every equivalent Hilbert norm $|\cdot|$ on the underlying vector space there exists $\delta > 0$ such that every closed infinite-dimensional subspace $Y \subset X$ has*

$$\inf_{a>0} \sup_{y \in Y \setminus \{0\}} \max \left\{ \frac{\|y\|}{a |y|}, \frac{a |y|}{\|y\|} \right\} \geq 1 + \delta.$$

Thus any counterexample must be uniformly distortive on every infinite-dimensional subspace, relative to every Hilbertian renorming.

Proof. This is the contrapositive of the theorem. $\square$

Limitations and Novelty Check

This packet does not solve the source question. It rules out only the almost-Hilbertian restriction case. The remaining case is the one where the equivalent Hilbert norm is genuinely distortive on every infinite-dimensional subspace. The common shortcut “ X is isomorphic to ℓ_2 , hence some subspace is arbitrarily close to Hilbert” is not valid for arbitrary equivalent Hilbert norms; distortive Hilbert renormings are a real obstruction.

The run’s lightweight indexes were searched for arXiv:2204.06834, the paper title, and the core terms from Question 3. No prior packet or attempt for this target was found. The source paper and de Rancourt’s partial result were inspected. Bounded web searches for the exact isometrically homogeneous Banach-space question did not reveal a later full answer during this pass.

References

References

- [1] M. Cuth, M. Dolezal, M. Doucha, and O. Kurka, *Polish spaces of Banach spaces. Complexity of isometry and isomorphism classes*, arXiv:2204.06834v1, 2022.
- [2] N. de Rancourt, *Spectral-free methods in the theory of hereditarily indecomposable Banach spaces*, Bull. Belg. Math. Soc. Simon Stevin 27 (2020), 775–787.
- [3] W. T. Gowers, *An infinite Ramsey theorem and some Banach-space dichotomies*, Ann. of Math. (2) 156 (2002), 797–833.
- [4] R. A. Komorowski and N. Tomczak-Jaegermann, *Banach spaces without local unconditional structure*, Israel J. Math. 89 (1995), 205–226.

Human Review Recommendation

Check that the stabilization hypothesis is recorded as an added hypothesis, not as a known consequence of $X \simeq \ell_2$. The proof itself is short: verify the transfer through the isometry $U : X \rightarrow Y$, the finite-dimensional distance-one compactness step, and the final parallelogram-law conclusion.